

HYBRID ELECTRIC VEHICLE

Engineering the Structural Logic of Dual-Power Propulsion Systems

A project-first certification designed to master Powertrain Integration, Battery Management Systems (BMS), and the structural engineering of multi-mode mobility.



THE EDLANX METHODOLOGY

OUR HEV ENGINEERING PHILOSOPHY

At Edlanx, we don't just teach "electric motors"; we cultivate a powertrain-architect mindset. Our program is a deep dive into the synergy between internal combustion and electric torque. We believe in Optimized Efficiency — moving beyond simple batteries to build robust, intelligent energy loops that ensure maximum range and performance in the global green mobility market.

THE FOUR PILLARS OF MASTERY

-  **Powertrain Logic** Master the core architecture of Series, Parallel, and Power-Split hybrid configurations.
-  **Energy Governance** Learn the structural logic of Battery Management Systems, SoC (State of Charge) estimation, and thermal safety.
-  **Control Engineering** Understand the engineering of Vehicle Control Units (VCU) and regenerative braking logic.
-  **System Integration** Bring your designs to life by mastering the structural link between high-voltage electronics and mechanical drivetrains.

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WHY CHOOSE EDLANX



01 Industry-Ready Curriculum

We move beyond basic electronics to bridge the gap between "circuits" and high-level vehicle engineering. Our Hybrid Electric Vehicle syllabus is a deep immersion into Dual-Power integration logic, Battery Management Systems (BMS), and the complex control strategies required by global automotive OEMs.

02 Real-Time Corporate Exposure

Step into a high-performance professional environment at our Bangalore Operational Center. At Edlanx, you won't just study motors; you will be treated as a Powertrain Engineer, working within real organizational frameworks to architect energy-management loops and manage high-voltage data flow.

03 Mentorship & Continuous Assessment

Your growth is guided by senior Automotive Architects and EV Systems Leads through structured mentoring and rigorous performance evaluations. We focus on developing your ability to solve complex torque-split hurdles and thermal-runaway risks with the precision required for production-ready vehicles.

04 Skill Mastery & Performance Incentives

Master industry-standard tools like MATLAB/Simulink, Ansys, and CAN Analyzers while learning essential professional competencies. We believe in experiential learning and reward innovative energy-recovery or powertrain breakthroughs with performance-based stipends, ensuring you remain driven toward technical mastery.

05 Global Career Readiness

We enhance your employability by focusing on the complete Vehicle Development Lifecycle, from initial plant modeling to final high-voltage diagnostic validation. Our career guidance includes technical resume engineering and mock "Design Gate Reviews" designed to help you secure elite roles in R&D centers globally.

06 Collaborative Engineering Community

Join a supportive culture built on transparency, integrity, and disciplined professional ethics. You will learn to lead a hybrid "sprint," collaborate across mechanical and electronic domains, and grow your network within an elite community of engineers, developers, and mentors.

07 Verified Professional Certifications

At Edlanx, we don't just issue certificates for attendance; we award them based on verified learning, real-world performance, and practical outcomes. Our credentials are proof of your industry-ready skills and are recognized by institutions and recruiters worldwide.

We provide four distinct levels of certification:

- Internship Completion Certificate: Validates your exposure to professional work culture, verified internship hours, and task-based mentorship.
 - Project Completion Certificate: Proof of your hands-on implementation and problem-solving skills in real-world industry projects.
 - Training Completion Certificate: Issued upon mastering our industry-aligned curriculum and passing skill-based assessments.
 - Certificate of Excellence: A prestigious award reserved for top-performing candidates who demonstrate an advanced understanding.
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Why This Journey Matters

-  **Portfolio-Ready** Every energy management strategy and BMS control algorithm you architect at Edlanx serves as a live, audit-ready demonstration of your ability to manage the deterministic complexity of modern zero-emission mobility.
-  **Global Network** Gain exclusive access to the Edlanx alumni community and industry mentors leading R&D teams at global Automotive OEMs, Tier-1 suppliers, and emerging EV startups.
-  **Career Excellence** We move beyond basic component knowledge to provide the technical resume engineering and systems-integration logic needed to lead the future of "Smart Mobility" and the global transition to electrified transport.

This course is a living ecosystem. We constantly update our curriculum to reflect the latest shifts in 800V architectures, Solid-State Battery integration, and AI-driven predictive energy management. Your success is our shared mission.

DIRECT MENTORSHIP



Engineering a reliable hybrid powertrain requires absolute mathematical and thermal precision. Whether you are optimizing a torque-split algorithm for a planetary gear set, designing a thermal management loop for a high-density battery pack, or architecting a CAN-FD communication matrix, our senior Automotive Architects are available to provide the direct, technical feedback needed to ensure your systems are production-ready.

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